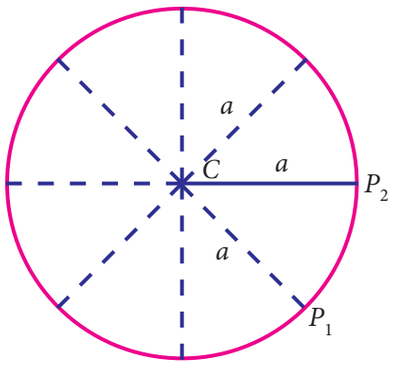


Fig. 3.1

For example,

(i) The locus of a point $P(x_1, y_1)$ whose distance from a fixed point $C(h, k)$ is constant, is a circle. The fixed point 'C' is called the centre.

(ii) The locus of a point whose distances from two points A and B are equal is the perpendicular bisector of the line segment AB.

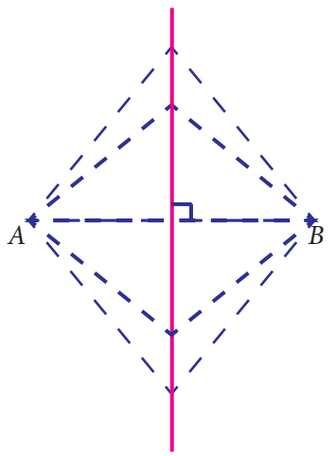


Fig. 3.2



Straight line is the locus of a point which moves in the same direction.

Example 3.1

A point in the plane moves so that its distance from the origin is thrice its distance from the y- axis. Find its locus.

Solution

Let $P(x_1, y_1)$ be any point on the locus and A be the foot of the perpendicular from $P(x_1, y_1)$ to the y-axis.

Given that OP $= 3$ AP

$$OP^2 = 9AP^2$$

$$(x_1 - 0)^2 + (y_1 - 0)^2 = 9x_1^2$$

$$x_1^2 + y_1^2 = 9x_1^2$$

$$8x_1^2 - y_1^2 = 0$$

$\therefore$ The locus of $P(x_1, y_1)$ is

$8x^2 - y^2 = 0$

Example 3.2

Find the locus of the point which is equidistant from (2, –3) and (3, –4).

Solution

Let $A(2, -3)$ and $B(3, -4)$ be the given points

Let $P(x_1, y_1)$ be any point on the locus.

Given that $PA = PB$.

$$PA^2 = PB^2$$

$$(x_1 - 2)^2 + (y_1 + 3)^2 = (x_1 - 3)^2 + (y_1 + 4)^2$$

$$x_1^2 - 4x_1 + 4 + y_1^2 + 6y_1 + 9 = x_1^2 - 6x_1 + 9 + y_1^2 + 8y_1 + 16$$

i.e., $2x_1 - 2y_1 - 12 = 0$

i.e., $x_1 - y_1 - 6 = 0$

The locus of $P(x_1, y_1)$ is $x - y - 6 = 0$.

Example 3.3

Find the locus of a point, so that the join of (–5, 1) and (3, 2) subtends a right angle at the moving point.

03_11th_BM-STAT_Ch-3-EM.indd 52 21-04-2020 12:14:39 PM